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Ceiling fan mounting kit

Abstract

A ceiling fan or similar air-moving device can include a motor for rotating one or more blades to drive a volume of air about a space. A mounting kit can be used to secure the ceiling fan to a structure. The mounting kit can be corded instead of hard-wired to the structure electrical supply, which provides the user with the freedom to mount the ceiling fan at any desired location, without being limited to an electrical junction box location.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims benefit of U.S. Provisional Patent Application No. 63/606,759, filed Dec. 6, 2023. This application is also related to U.S. patent application Ser. No. 29,925,499, filed Jan. 24, 2024, and U.S. patent application Ser. No. 29/933,830, filed Mar. 21, 2024, all of which are hereby incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

(1) This application is directed to a ceiling fan, and more specifically, to a kit for mounting a ceiling fan to a structure without a hard-wired electrical connection.

BACKGROUND

(2) Ceiling fans are machines traditionally suspended from a structure for moving a volume of air about an area. The ceiling fan includes a motor, with a rotor and a stator, suspended from and electrically coupled to the structure. A set of blades mount to the rotor such that the blades are rotatably driven by the rotor and can be provided at an angled orientation to move volume of air about the area.

(3) The ceiling fan is secured to the structure by a bracket, which is often referred to as a ceiling plate or mounting bracket, which has a pass-through opening through which passes the electrical leads of one or both of the structure electrical supply or the fan. The electrical leads from the structure and fan can then be hard wired together using traditional wire nuts.

BRIEF DESCRIPTION

(4) In one aspect, the disclosure relates to a ceiling fan mounting kit for a non-hard-wired electrical connection, the kit comprising: a mounting bracket having fastener openings through which fasteners can pass to secure the mounting bracket to a structure; a strain relief for an electrical cord; and an electrical cord having a plug on one end.

(5) In another aspect, the disclosure relates to A ceiling fan mounting kit for a non-hard-wired electrical connection, the kit comprising: a mounting bracket having fastener openings through which fasteners can pass to secure the mounting bracket to a structure; an electrical cord having a plug on one end and a set of wire leads on another end; a strain relief securing the electrical cord to the mounting bracket; and a wire connector carried by the mounting bracket and receiving the set of wire leads from the electrical cord.

(6) In yet another aspect, the disclosure relates to a ceiling fan mounting assembly for securing a ceiling fan to a structure, the ceiling fan mounting assembly comprising: a mounting bracket having a base and a peripheral wall, with the base having fastener openings through which fasteners can pass to secure the mounting bracket to

the structure, the peripheral wall having a cord notch; a wire connector carried by the base; an electrical cord having a plug on one end and a set of wire leads on another end, wherein the cord passes through the cord notch and the wire leads are received in the wire connector; and a strain relief carried by the base and securing the electrical cord to the mounting bracket.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the drawings:
- (2) FIG. 1 is a perspective view of a structure with a ceiling fan and a mounting kit securing the ceiling fan to the structure.
- (3) FIG. 2 is a partial exploded view of the mounting kit of FIG. 1.
- (4) FIG. 3 is a bottom view of a mounting bracket from the mounting kit of FIG. 2.
- (5) FIG. 4 is a bottom view of a gasket from the mounting kit of FIG. 2.
- (6) FIG. 5 is a view of a kit with options for the mounting kit of FIG. 2.
- (7) FIG. 6 is a perspective view of a second embodiment of the mounting kit.
- (8) FIG. 7 is an exploded view of the mounting bracket for the mounting kit of FIG. 6.

DETAILED DESCRIPTION

(9) The disclosure is related to a ceiling fan and mounting kit, which can be used, for example, in residential and commercial applications. Such applications can be indoors, outdoors, or both. While this description is primarily directed toward a residential ceiling fan, it is also applicable to any environment utilizing fans or for cooling areas utilizing air movement.

(10) As used herein, the term “set” or a “set” of elements can be any number of elements, including only one. All directional references (e.g., radial, axial, proximal, distal, upper, lower, upward, downward, left, right, lateral, front, back, top, bottom, above, below, vertical, horizontal, clockwise, counterclockwise, upstream, downstream, forward, aft, etc.) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of aspects of the disclosure described herein. Connection references (e.g., attached, coupled, connected, and joined) are to be construed broadly and can include intermediate members between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to one another. The exemplary drawings are for purposes of illustration only and the dimensions, positions, order and relative sizes reflected in the drawings attached hereto can vary.

(11) Referring now to FIG. 1, a ceiling fan **10** is suspended from a structure **12** by a mounting kit **14**. In non-limiting examples, the ceiling fan **10** can include one or more ceiling fan components including a downrod **16**, a hanger ball **20**, a motor housing **22** at least partially encasing a motor (not shown) having a rotor and a stator, a light kit **24**, and a set of blades **26**, which can have optional blade irons (not shown), depending on the ceiling fan configuration. In additional non-limiting examples, the ceiling fan **10** can include one or more of a controller, a wireless receiver, a light glass, a light cage, a spindle, a finial, a switch housing, blade forks, blade tips or blade caps, or other ceiling fan components.

(12) The structure **12** can be a ceiling, for example, from which the ceiling fan **10** is suspended. It should be understood that the structure **12** is schematically shown and is by way of example only, and can include any suitable building, structure, home, business, or other environment wherein moving air with a ceiling fan is suitable or desirable.

(13) While not shown in the drawings, a controller can be electrically coupled to the electrical supply to control operation of the ceiling fan **10**. Alternatively, the controller can be wirelessly or communicatively coupled to the ceiling fan **10**, configured to control operation of the ceiling fan **10** remotely, without a dedicated connection. Non-limiting examples of controls for the ceiling fan **10** can include fan speed, fan direction, or light operation. Furthermore, a separate wireless controller, alone or in addition to the wired controller, can be communicatively coupled to a controller or a wireless receiver in the ceiling fan **10** to control operation of the ceiling fan **10**. It is further contemplated in one alternative example that the ceiling fan **10** be operated by the wireless controller alone and is not operably coupled with the wired controller.

(14) Referring to FIG. 2, the mounting kit **14** is illustrated partially exploded to illustrate the different components of the mounting kit **14**, which are illustrated as including: mounting bracket **30**, hanger bracket **32**, wire connector **34**, strain relief **36**, cord **38**, gasket **40**, and canopy **42**. For case of description, the canopy **42** is described as part of the mounting kit **14**. However, the canopy **42** can be considered as part of the ceiling fan **10**

and is optional to the mounting kit **14**.

(15) The mounting bracket **30** is secured to the structure **12**, such as by suitable fasteners (screws, bolts, etc) that are secured directly to the structure **12** or the framing in a wood-frame structure. Additionally, suitable anchors (drywall, concrete, etc.) can be affixed to the structure **12** and the mounting bracket **30** is secured to the anchors. Other methods of securing the mounting bracket **30** to the structure **12** are contemplated, and the particular method is not limiting. The mounting bracket **30** forms the primary physical connection between the structure **12** and the ceiling fan **10**. As illustrated, the mounting bracket **30** also forms the primary structure for carrying the other components of the mounting kit **14**.

(16) The hanger bracket **32** is secured to the mounting bracket **30**, such as, for example, by the illustrated fasteners **44**, and has a ball seat **46** for receiving the hanger ball **20** to secure the motor housing **22**, via the downrod **16**, to the structure **12**. The hanger bracket **32** further has tabs with tab openings **48** for use in securing the canopy **42**.

(17) The wire connector **34** is secured to the mounting bracket **30**, such as by the illustrated clamp **50**, which can be secured to the mounting bracket **30** by suitable fasteners, such as bolts or screws, for example. The wire connector **34** has a first set of wire lead receptacles that receive wire leads from the cord **38** and at least a second set of wire lead receptacles that receive wire leads from one or both of the motor or controller. Internally of the wire connector **34** the first and second sets of the wire lead receptacles are electrically connected. In this way, the wire connector **34** forms a common junction for the household power supply and the elements of the ceiling fan **10** requiring power.

(18) The strain relief **36** is secured to the mounting bracket **30**, such as, for example, by the illustrated fastener **52**, and comprises a clamp band **54**, which upon the tightening of the fastener **52** tightens the clamp band **54**. A band of compressible material **56** can be carried by the interior of the clamp band **54** and provide a slightly compliant and non-sharp interface with the cord **38**. The compressible material **56** can also form a liquid seal.

(19) The cord **38** terminates on one end in a plug **58** and the other end in a set of wire leads **60**, which are received within the first set of receptacles for the wire connector **34** after the cord **38** is passed through the clamp band **54** of the strain relief **36**. The cord **38** is secured to the mounting bracket **30** by the clamp **50** band, which tightens around the cord **38** as well as the frictional holding of the wire connector **34**.

(20) The canopy **42** is a decorative cover that visually hides most of the mounting kit components carried by the mounting bracket **30**. The canopy **42** has an opening **62** through which the hanger ball **20** and downrod **16** can pass as well as fastener openings **64**. Suitable fasteners (not illustrated), such as screws or bolts, can pass through the fastener openings **64** and into the tab openings **48** of the hanger bracket **32** to secure the canopy **42** to the hanger bracket **32**. The canopy **42** also defines an interior volume **66** accessible through an open upper end **68**, whereby the canopy **42** can be pushed up toward the mounting bracket **30** such that the components of the mounting kit **14** carried by the mounting bracket **30** are received within the interior volume **66**.

(21) Referring to FIG. **3**, the mounting bracket **30** is shown in greater detail and with the removal of all the other components from the mounting kit **14**. The mounting bracket **30** comprises a base **70**, with an optional peripheral wall **72**, which bounds an interior of the base **70**. Various embossments are formed in the base to case the mounting of the different mounting kit components. For example, hanger bracket embossments **74** are provided and have openings **76** that receive fasteners, such as screws or bolts, for securing the hanger bracket **32** to the base **70**. A strain relief embossment **78** has an opening **80** that also receives the fastener. A cord notch **82** is formed in the peripheral wall **72** adjacent the strain relief embossment **78** and provides an open area in the peripheral wall **72** through which the cord **38** can pass. The location of the cord notch **82** is axially aligned with the location of the clamp **50** band to case the passing of the cord **38**.

(22) A cord gasket **83** is received within the cord notch **82** and has an opening **85** through which the cord **38** passes. The cord gasket **83** forms a liquid seal around the chord **38** and between the cord **38** and the peripheral wall **72**, and, optionally, the base **70**. The cord gasket **83** can have a ring **87** forming the opening **85**, with the ring **87** received within the cord notch **82**. The ring **87** can have a split **91** to case the insertion of the cord **38** through the opening **85**. A backwall **93** bears against an inner surface of the peripheral wall **72** while supporting the ring **87**. The backwall **93** functions as a stop to limit the insertion of the ring **87** into the cord notch **82** as well as retard the removal of the cord gasket **83** through the cord notch **82**.

(23) Various openings are formed in the base **70** to aid in the securing of the mounting bracket **30** to the structure. As illustrated, there are multiple slots **84**, which provide openings through which fasteners, such as screws or bolts, can pass through and be affixed into the structure. The multiple slots could be a single slot, instead of the multiple discrete slots. The slots, while illustrated as being arcuate, could take on different shapes, such as linear. Other openings **86** are provided for mounting the wire connector **34** by suitable fasteners.

(24) Notably, the base **70** does not have an opening for the passage of electrical wire leads from the householder

power supply, unlike prior mounting brackets. Prior mounting brackets were designed to be located at an electrical junction box, typically in the ceiling. The prior mounting bracket would be secured directly to the junction box and the wire leads from the household power supply would pass from the junction box through the large central opening of the mounting bracket **30**. Such a configuration is not needed with the described mounting bracket **30** since the cord **38** is used, instead of the hard-wired, wire leads from the household power supply, to connect to the household power supply. The use of the cord **38** in place of the hard-wired, household wire leads frees the mounting kit **14** from being located where an electrical junction box is located. This freedom provides the consumer with the benefit of locating the mounting kit **14** anywhere they like on the structure and not just where a junction box is located. Previously, consumers would be forced to place the ceiling fan **10** where a junction box was present or have an electrician install a new junction box, at substantial cost, where it was desired to place the ceiling fan **10**. With the corded mounting kit **14**, the consumer is free to place the ceiling fan **10** anywhere they like, without consideration of the prewired junction box. Such freedom is beneficial for both indoor and outdoor ceiling fan **10** mountings, but especially for outdoor mountings where junction boxes are typically not present, but an outdoor electrical outlet, which can receive the plug end of the cord **38** is present.

(25) FIG. **4** illustrates the gasket **40**, which is optional for the mounting kit. The gasket **40** is primarily intended to prevent water from entering the openings in the base of the mounting bracket **30**. As such, the gasket **40** is primarily intended for use in outdoor installations. However, there is no reason why the gasket **40** cannot be used in indoor installations. The gasket **40** has a gasket base **90** and gasket peripheral wall **92**, similar to the similar structures of the mounting bracket **30**. Advantageously, the gasket base **90** and gasket peripheral wall **92** of the gasket **40** can be complementary with those of the mounting bracket **30** such that they form a water seal therewith. The gasket base **90** and gasket peripheral wall **92** define a gasket recess **94** that is sized to receive the mounting bracket **30** such that the gasket base **90** abuts the base **70** and the gasket peripheral wall **92** abuts the peripheral wall **72**. The gasket peripheral wall **92** is illustrated as being shorter than the peripheral wall **72**, but that need not be the case. The gasket peripheral wall **92** improves the water scaling function by wrapping around the mounting bracket **30** to prevent water from easily entering the interface between the gasket **40** and the mounting bracket **30**.

(26) FIG. **5** illustrates the various contemplated commercially available kits for the mounting kit. In the simplest, most bare bones form, the mounting kit would be commercially sold in a kit **100** that includes just the mounting bracket **30**, cord **38**, and strain relief **36**, which will be referenced herein as the “base kit”. With such a kit, it would be contemplated that the consumer would use an already existing hanger bracket **32** and canopy **42** from the ceiling fan **10**. It is further contemplated that the consumer would hard wire the cord **38** wire leads to the motor electrical leads using tradition wire nuts.

(27) In a first variation of the base kit, the gasket **40** can be added, especially when an outdoor installation is contemplated. In a second variation, the wire connector **34** can be included with the base kit and/or with the first variation. In a third variation, the hanger bracket **32** can be included with the base kit, first variation, or third variation. In many cases, the consumer could use the hanger bracket **32** from the ceiling fan **10**. However, many consumers may not want to disassembly the hanger bracket **32** from the ceiling fan **10** and assemble it to the mounting bracket **30**, and/or the prior hanger bracket **32** may not be compatible with the mounting bracket **30**. Therefore, it will be more convenient for many consumers to provide the hanger bracket **32** with the commercially available kit, especially pre-assembled to the mounting bracket **30**.

(28) In all of the different commercially available kits, the different components of the mounting kit can be unassembled, pre-assembled, or a portion of them pre-assembled. The greater the degree of pre-assembly, the easier it will be on the consumer.

(29) Referring still to FIG. **5**, all of the kits can include ceiling fan-specific instructions **98** to guide the installer, who is contemplated to be a consumer of the fan, through one or more of the following exemplary installation steps of: locating the mounting bracket **30** relative to the structure **12**, securing the mounting bracket **30** to the structure, securing the cord to the strain relief **36**, connecting the wire leads **60** to the wire connector **34**, securing the hanger ball **20** to the hanger ball seat **46**, and affixing canopy **42** to the hanger bracket **32**.

Depending on the variation of the kit, the instructions **98** may include securing the hanger bracket **32** to the mounting bracket **30** or securing the wire connector **34** to the mounting bracket **30**. The instructions **98** can also include guiding the installer on a suitable electrical outlet in which the plug **58**. The instructions **98** can be a separate printed sheet or booklet, which is included with the packaging of the kit. The instructions **98** can also be printed on the packaging. For example, the packaging can include a box containing the kit and the printed sheet or booklet can be provided in the box or printed on a surface of the box. In some cases, the packaging can be a cardboard backing with a plastic wrap or blister pack, with the components of the kit located between the

cardboard backing and the plastic wrap. The printed sheet or booklet can be sandwiched between the cardboard backing and plastic sheet as well or the instructions can be printed on the cardboard backing.

(30) While it is contemplated the consumer will use the canopy **42** that comes with the ceiling fan **10**, another variation to all of the previously described mounting kits, is the additional of a canopy **42**.

(31) Referring to FIG. **6**, a second embodiment mounting kit **214** is illustrated. The mounting kit **214** is similar to the mounting kit **14**. Therefore, like parts will be identified with like numerals increased by **200**. Also, the prior description for mounting kit **14** is applicable to mounting kit **214**, including all of the variants. Therefore, the following description of the mounting kit **214** will highlight the differences with respect to mounting kit **14**.

(32) The mounting kit **214** is illustrated as including: mounting bracket **230**, wire connector **234**, strain relief **236**, cord **238**, and gasket **240**. The mounting kit **214** is illustrated without a canopy or a hanger bracket to simplify the description. The canopy **42** and the hanger bracket **32** can be used with the mounting kit **214** and their prior description is applicable to the mounting kit **214**. The canopy and hanger bracket are also optional for the mounting kit **214**.

(33) The mounting kit **214** further includes cord clips **300** and corresponding fasteners **302**. While fourteen cord clips **300** are shown, and the corresponding number of fasteners **302**, any desired number can be provided. While the chord clips **300** are not shown with the mounting kit **14**, they can be used with the mounting kit **14**. Any suitable cord clips **300** and any fasteners **302** can be used. The cord clips **300** are secured to the structure **12** and help hold the cord **238** in place against the structure **12**. The fasteners **302** are illustrated as screws. However, they could also include an anchor, such as a drywall anchor or concrete anchor, for example, in which the screw is received. Depending on the structure **12**, the cord clips **300** could be attached by an adhesive instead of a fastener.

(34) The mounting kit **214** further includes a cord gasket **283** that forms a liquid seal between the cord **238** and a cord notch **282** in the mounting bracket **230**. The strain relief **236** is different from the strain relief **36** in that it is structurally different and it is inset further interiorly of the cord notch **282**.

(35) The mounting bracket **230**, strain relief **236**, and cord gasket **283** are better seen in FIG. **7**, which shows the strain relief **236** and cord gasket **283** partially exploded from the mounting bracket **230**. As with the mounting bracket **30**, the mounting bracket **230** comprises a base **270**, with an optional peripheral wall **272**, which bounds an interior of the base **70**. The peripheral wall **272** includes a drip edge **312**. The drip edge **312** causes liquid running down the exterior of the peripheral wall **272** to drip off the mounting bracket **230** instead of the following the contour of the peripheral wall **272** into the interior of the mounting bracket **230**. The drip edge **312** can be added to the mounting bracket **30**. During operation, it should be noted that the mounting bracket **230** will be oriented 180 degrees (face down) from what is seen in FIG. **7** (face up). In the operative orientation, liquid will flow along the outside of the peripheral wall **272** and down to the lowest extent of the drip edge **312**, where the liquid will not be able to turn the corner formed by the drip edge **312**, and the liquid will drip off the drip edge **312**.

(36) Various embossments are formed in the base **270** to ease the mounting of the different mounting kit components. For example, hanger bracket embossments **274** are provided and have openings **276** that receive fasteners, such as screws or bolts, for securing the hanger bracket **32** to the base **270**. A strain relief embossment **278** has an opening **280** that also receives the fastener **281**. The cord notch **282** is formed in the peripheral wall **272** and provides an open area in the peripheral wall **272** through which the cord **238** can pass.

(37) Various openings are formed in the base **270** to aid in the securing of the mounting bracket **230** to the structure **12**. As illustrated, there are multiple slots **284**, which provide openings through which fasteners, such as screws or bolts, can pass through and be affixed into the structure **12**. The multiple slots could be a single slot, instead of the multiple discrete slots. The slots, while illustrated as being arcuate, could take on different shapes, such as linear. Other openings **286** are provided for mounting the wire connector **234** by suitable fasteners.

(38) The base **270** also includes a recess **314** sized to receive the wire connector **234**. The recess **314** is shaped to complement and index the wire connector **234** and, therefore, simplifies the locating and alignment of the wire connector **234** to the base **270**. The base **270** further includes indicia associated with the wire connector **234**. The indica indicates the orientation and wiring of the wire connector **234** in terms of the three household wire leads: Live **316**, Ground **318**, and Neutral **320**. The indica helps an installer when the household wiring has a different color sheath than is currently found in the industry. As with the base **70**, the base **270** does not have an opening for the passage of electrical wire leads from the householder power supply for the same purposes as the base **70**.

(39) The strain relief **236** comprises a clamp band **254** having one end, with a tab **330**, that is received in a clamp band recess **332** in the base **270** and another end that has a fastener opening **334** that overlies the opening

280 in the strain relief embossment **274**, which upon the receipt of the fastener **252** in the aligned openings **334**, **280**, the clamp band **254** is secured to the strain relief embossment **274**. The clamp band **254** includes an arcuate portion **336** that conforms to the contour of the cord **238**.

(40) One or more teeth **338** protrude from the base **270** and underlie the clamp band **254**, especially the arcuate portion **336**, such that the cord **238** is pressed into the teeth **338** as the clamp band **254** is tightened. The teeth **338** help hold the cord **238** in place.

(41) The cord gasket **283** comprises a ring **287** depending from a tab **342**. The peripheral wall **272** includes a pair of fingers **344** on opposite sides of the cord notch **282**. A ring slot **346** is formed in the fingers **344**, and the base **270**, to receive and hold the ring **287**. The tab **342** forms a seal across the terminal ends of the fingers **344**. The tab **342** also functions to continue the drip edge across the cord notch **282**.

(42) The second embodiment mounting kit **214** can be arranged in a kit in the same manner as the first embodiment, with or without, the additional kit variations described for the second embodiment. Similarly, the first embodiment can be arranged in a kit according to the variations for the second embodiment. For example, while the first embodiment is not shown with the cord clips of the second embodiment, the cord clips can be added to the first embodiment. Further, the seal edge and/or the cord gasket of the second embodiment can be used with the first embodiment. Similarly, either of the embodiments can use the strain relief from the other embodiment. These examples are not meant to be limiting on the combination of variations from one embodiment to the other embodiment. Instead, these examples are just that, possible examples, to understand what is meant by adding a variation of one embodiment to the other and how the variations can be mixed and matched between the disclosed embodiments in making a kit.

(43) Also, in making the kits, multiple and/or duplicate parts can be included, and the parts can cross embodiments. For example, both strain reliefs can be placed in the same kit. Alternatively, multiple of the same strain relief can be placed in the same kit. This use of multiple/duplicate parts applies to all of the parts for both embodiments.

(44) Throughout the disclosure, suitable fasteners are described for securing different components. Such fastener can be any type of fastener. In many cases the suitable fastener is a screw or a bolt. If a bolt, then the opening that the bolt is threaded into will be a tapped opening having complementary threads for receipt of the bolt. The fasteners may be used alone or in combination with a suitable anchor, especially when the fastener attaches to the structure. Suitable anchors can be material dependent, such as drywall anchors or concrete anchors.

(45) To the extent not already described, the different features and structures of the various features can be used in combination as desired. That one feature is not illustrated in all of the aspects of the disclosure is not meant to be construed that it cannot be but is done for brevity of description. Thus, the various features of the different aspects described herein can be mixed and matched as desired to form new features or aspects thereof, whether or not the new aspects or features are expressly described. All combinations or permutations of features described herein are covered by this disclosure.

(46) Both of the strain reliefs described and illustrated are advantageous in that they permit the installer to remove, such as by cutting, any excess cord, strip the end of the wires, and connect the stripped end to the wire connector. The strain relief can then be tightened on the shortened cord. In this way, the strain relief aids the installer in removing any excess cord to provide a more custom installation.

(47) This written description uses examples to detail the aspects described herein, including the best mode, and to enable any person skilled in the art to practice the aspects described herein, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the aspects described herein are defined by the claims, and can include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

(48) Further aspects are provided by the subject matter of the following clauses:

(49) A ceiling fan mounting kit for a non-hard-wired electrical connection, the kit comprising: a mounting bracket having fastener openings through which fasteners can pass to secure the mounting bracket to a structure; a strain relief for an electrical cord; and an electrical cord having a plug on one end.

(50) The kit according to any of the other clauses further comprising a wire connector.

(51) The kit according to any of the other clauses wherein the wire connector has a first set of wire lead receptacles for the cord and a second set of wire lead connectors for the ceiling fan.

(52) The kit according to any of the other clauses wherein another end of the cord has wire leads for receipt within the first set of wire lead receptacles.

(53) The kit according to any of the other clauses further comprising a hanger bracket having a hanger ball seat.

- (54) The kit according to any of the other clauses wherein the hanger bracket comprises canopy mounting tabs.
- (55) The kit according to any of the other clauses further comprising a canopy.
- (56) The kit according to any of the other clauses wherein the wire connector is carried by the mounting bracket.
- (57) The kit according to any of the other clauses wherein the cord strain relief is carried by the mounting bracket.
- (58) The kit according to any of the other clauses further comprising a hanger bracket.
- (59) The kit according to any of the other clauses further comprising a gasket for use between the mounting bracket and structure.
- (60) The kit according to any of the other clauses wherein the gasket comprises a recess for receipt of the mounting bracket.
- (61) A ceiling fan mounting kit for a non-hard-wired electrical connection, the kit comprising: a mounting bracket having fastener openings through which fasteners can pass to secure the mounting bracket to a structure; an electrical cord having a plug on one end and a set of wire leads on another end; a strain relief securing the electrical cord to the mounting bracket; and a wire connector carried by the mounting bracket and receiving the set of wire leads from the electrical cord.
- (62) The kit of any of the other clauses further comprising a gasket to seal the mounting bracket with the structure.
- (63) The kit of any of the other clauses wherein the gasket comprises a recess that receives at least a portion of the mounting bracket.
- (64) The kit of any of the other clauses wherein the gasket and mounting bracket have complementary planar surfaces.
- (65) The kit of any of the other clauses wherein the complementary planar surfaces have complementary fastener openings.
- (66) The kit of any of the other clauses wherein the gasket and mounting bracket have complementary peripheral walls.
- (67) The kit of any of the other clauses wherein the mounting bracket peripheral wall is radially interior of the gasket peripheral wall.
- (68) The kit of any of the other clauses wherein the complementary peripheral walls are in contact.
- (69) A ceiling fan mounting assembly for securing a ceiling fan to a structure, the ceiling fan mounting assembly comprising: a mounting bracket having a base and a peripheral wall, with the base having fastener openings through which fasteners can pass to secure the mounting bracket to the structure, the peripheral wall having a cord notch; a wire connector carried by the base; an electrical cord having a plug on one end and a set of wire leads on another end, wherein the cord passes through the cord notch and the wire leads are received in the wire connector; and a strain relief carried by the base and securing the electrical cord to the mounting bracket.
- (70) The ceiling fan mounting assembly of any of the other clauses wherein the peripheral wall further comprises a drip edge.
- (71) The ceiling fan mounting assembly of any of the other clauses wherein the drip edge extends continuously about the peripheral wall, except for the cord notch.
- (72) The ceiling fan mounting assembly of any of the other clauses further comprising a cord gasket providing a drip edge across the cord notch.
- (73) The ceiling fan mounting assembly of any of the other clauses wherein the cord gasket forms a liquid seal within the cord notch.
- (74) The ceiling fan mounting assembly of any of the other clauses wherein the liquid seal extends between the cord and the mounting bracket.
- (75) The ceiling fan mounting assembly of any of the other clauses wherein the cord gasket defines a cord opening through which the cord passes.
- (76) The ceiling fan mounting assembly of any of the other clauses wherein the cord gasket defines a ring forming the cord opening.
- (77) The ceiling fan mounting assembly of any of the other clauses wherein the cord gasket further comprises a tab abutting the peripheral wall and the ring depending from the tab.
- (78) The ceiling fan mounting assembly of any of the other clauses wherein the tab continues the drip edge across the cord notch.
- (79) The ceiling fan mounting assembly of any of the other clauses wherein the peripheral wall defines a ring slot that receives the ring.
- (80) The ceiling fan mounting assembly of any of the other clauses further comprising a cord gasket forming a liquid seal between the cord and the mounting bracket.

- (81) The ceiling fan mounting assembly of any of the other clauses wherein the cord gasket forms a liquid seal within the cord notch.
- (82) The ceiling fan mounting assembly of any of the other clauses wherein the cord gasket defines a cord opening through which the cord passes.
- (83) The ceiling fan mounting assembly of any of the other clauses wherein the cord gasket defines a ring forming the cord opening.
- (84) The ceiling fan mounting assembly of any of the other clauses wherein the peripheral wall defines a ring slot that receives the ring.
- (85) The ceiling fan mounting assembly of any of the other clauses wherein the peripheral wall has two spaced fingers in which at least a portion of the ring slot is formed.
- (86) The ceiling fan mounting assembly of any of the other clauses wherein the cord gasket further comprises a tab abutting the peripheral wall.
- (87) The ceiling fan mounting assembly of any of the other clauses wherein the tab spans the cord notch.
- (88) The ceiling fan mounting assembly of any of the other clauses wherein the tab forms a drip edge over the cord notch.
- (89) The ceiling fan mounting assembly of any of the other clauses wherein the strain relief further comprises a clamp band securing the cord to the mounting bracket.
- (90) The ceiling fan mounting assembly of any of the other clauses wherein the strain relief further comprises compressible material surrounding at least a portion of the cord.
- (91) The ceiling fan mounting assembly of any of the other clauses wherein the compressible material circumscribes at least a portion of the cord.
- (92) The ceiling fan mounting assembly of any of the other clauses wherein the clamp band wraps around the compressible material.
- (93) The ceiling fan mounting assembly of any of the other clauses wherein the strain relief further comprises at least one tooth extending from the mounting bracket and located below the clamp band.
- (94) The ceiling fan mounting assembly of any of the other clauses wherein the at least one tooth comprises multiple teeth.
- (95) The ceiling fan mounting assembly of any of the other clauses wherein the clamp band comprises a portion conforming to the shape of the cord.
- (96) The ceiling fan mounting assembly of any of the other clauses wherein the clamp band comprises a tab and the mounting bracket comprises a slot receiving the tab.

Claims

1. A ceiling fan mounting kit for a non-hard-wired electrical connection, the kit comprising: a mounting bracket having fastener openings through which fasteners can pass to secure the mounting bracket to a structure; a strain relief carried by the mounting bracket; an electrical cord coupled to the strain relief and having a plug on one end and wire leads on another end; and a hanger bracket having a hanger ball seat; wherein the mounting bracket further comprises a peripheral wall with a cord notch through which the electrical cord passes; wherein the strain relief is aligned with the cord notch; and wherein the wire leads are located interior of the peripheral wall and the plug is exterior of the peripheral wall.
2. The kit according to claim 1, further comprising a wire connector.
3. The kit according to claim 2, wherein the wire connector has a first set of wire lead receptacles for the cord and a second set of wire lead connectors for a ceiling fan.
4. The kit according to claim 3, wherein the wire leads are for receipt within the first set of wire lead receptacles.
5. The kit according to claim 4, wherein the hanger bracket comprises canopy mounting tabs.
6. The kit according to claim 5, further comprising a canopy.
7. The kit according to claim 5, wherein the wire connector is carried by the mounting bracket.
8. The kit according to claim 3, further comprising a gasket for use between the mounting bracket and the structure.
9. The kit according to claim 8, wherein the gasket comprises a recess for receipt of the mounting bracket.
10. The kit of claim 1, wherein the mounting bracket further comprises a base in which the fastener openings are located and the base does not have an opening for passage of electrical wire leads from a household power supply.
11. The kit of claim 1, wherein the strain relief comprises a clamp that clamps against the electrical cord.

12. A ceiling fan mounting kit for a non-hard-wired electrical connection, the kit comprising: a mounting bracket having fastener openings through which fasteners can pass to secure the mounting bracket to a structure; an electrical cord having a plug on one end and a set of wire leads on another end; a strain relief securing the electrical cord to the mounting bracket; a wire connector carried by the mounting bracket and receiving the set of wire leads from the electrical cord; and a hanger bracket having a hanger ball seat; wherein the mounting bracket further comprises a peripheral wall with a cord notch through which the electrical cord passes; wherein the strain relief is aligned with the cord notch; and wherein the wire leads are located interior of the peripheral wall and the plug is exterior of the peripheral wall.

13. The kit of claim 12, further comprising a gasket to seal the mounting bracket with the structure.

14. The kit of claim 13, wherein the gasket comprises a recess that receives at least a portion of the mounting bracket.

15. The kit of claim 14, wherein the gasket and the mounting bracket have complementary planar surfaces.

16. The kit of claim 15, wherein the complementary planar surfaces have complementary fastener openings.

17. The kit of claim 16, wherein the gasket has a peripheral wall complementary to the peripheral wall of the mounting bracket.

18. The kit of claim 17, wherein the peripheral wall of the mounting bracket is radially interior of the peripheral wall of the gasket.

19. The kit of claim 18, wherein the complementary peripheral walls are in contact.

20. The kit of claim 12, wherein the mounting bracket further comprises a base in which the fastener openings are located and the base does not have an opening for passage of electrical wire leads from a household power supply.

21. The kit of claim 12, wherein the strain relief comprises a clamp that clamps against the electrical cord.
